CONVERSATION-STATE BASED APPROACH

Assignment on

Assignment Report submitted in partial fulfilment of the requirements for the Course of

DSA0308 – Natural Language Processing

to the award of the degree of

BACHELOR OF ENGINEERING

IN

CSE - AI

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STATISTICAL CLASSIFICATION APPROACH

1. Problem Statement and Problem Formulation

Problem Statement

E-commerce customer-service systems must identify what a customer is trying to accomplish from natural-language messages. Customers may describe the same requirement using different vocabulary and grammatical forms. For example, a delivery-related request may contain words such as *arrive*, *arrived*, *arrival*, *delivery*, or *delivered*.

A chatbot based purely on exact keyword matching may treat these expressions as unrelated or may assign an incorrect intent when a competing word such as *cancel* occurs in the conversation.

The proposed system therefore uses a **statistical NLP approach based on TF-IDF features and n-grams** to classify customer messages. Linguistic processing and contextual correction are then applied to the classifier output so that explicit negation and conversational information can modify an otherwise misleading prediction.

The assignment requires the final classification to use multiple NLP levels rather than keyword frequency alone.

Problem Formulation

Input:
A sequence of customer messages from an e-commerce conversation.

Output:

- Predicted customer intent
- Intent probability/confidence
- Important entities
- Linguistic evidence
- Contextual corrections
- Retrieved information
- Generated response
- Tamil translation

The predefined intent categories include Order Tracking, Order Cancellation, Product Information, Return Request, Refund Request, Payment Problem, Delivery Information, Account Problem, Complaint and General Inquiry.

2. Objective and Expected Outcomes

Objectives

The main objective is to develop a statistical intent-classification system that can recognize customer requirements from different linguistic expressions.

The specific objectives are:

1. Preprocess customer conversations.
2. Reduce morphological variations through stemming.
3. Generate unigram, bigram and trigram features.
4. Convert textual input into numerical TF-IDF vectors.
5. Train a statistical classifier for intent prediction.
6. Use POS and syntactic information as supporting evidence.
7. Detect negation explicitly.
8. Identify e-commerce entities.
9. Resolve context-dependent references.
10. Incorporate discourse information.
11. Apply contextual correction to the statistical prediction.
12. Retrieve information associated with the selected intent.
13. Generate a natural-language response.
14. Translate the response into Tamil.

The assignment specifically requires unigram, bigram and trigram representations, POS tagging, negation handling, semantic processing, reference resolution and final intent classification.

Expected Outcomes

The system is expected to classify the given conversation as:

Final Intent: Order Tracking / Delivery Information

The expected result is obtained because the statistical model recognizes delivery-related patterns, while contextual correction identifies that cancellation has been explicitly rejected.

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3. Requirements, Constraints and Assumptions

Requirements

Functional Requirements

The proposed system should:

- Accept customer conversations.
- Clean and tokenize the text.
- Perform stemming.
- Generate TF-IDF representations.
- Include unigram, bigram and trigram features.
- Perform POS tagging.
- Analyze sentence structure.
- Detect negation.
- Identify entities.
- Resolve references.
- Analyze conversational markers.
- Predict an intent using a trained classifier.
- Apply contextual correction.
- Retrieve appropriate information.
- Generate a response.
- Translate the response into Tamil.

These requirements follow the functional requirements specified in the assignment.

Technical Requirements

Component	**Proposed Tool**
Programming Language	Python
Text Processing	NLTK
Stemming	PorterStemmer
Vectorization	Scikit-learn TF-IDF
N-grams	Scikit-learn
Classifier	Logistic Regression / Linear SVM

Component	Proposed Tool
POS Tagging	NLTK
CFG Parsing	NLTK
Semantic Processing	Python/NLP libraries
Retrieval	Python dictionary
Translation	Suitable translation library/API

The assignment permits NLTK, spaCy, Python regular expressions, Scikit-learn and other relevant NLP tools.

Constraints

The classifier cannot operate as a simple keyword-counting system.

The system must:

- use multiple linguistic signals;
- account for negation;
- preserve conversation context;
- resolve references;
- distinguish entities from intents;
- provide intermediate outputs;
- avoid contradictory responses;
- justify the final prediction.

Assumptions

1. A labelled training dataset containing e-commerce intents is available.
2. Each training example is assigned one primary intent.
3. Customer conversations are available in text format.
4. The classifier is trained before chatbot execution.
5. A predefined intent vocabulary is maintained.
6. The system can access a basic information-retrieval store.
7. Translation is performed after generating the English response.

4. Application of Relevant Course Knowledge / Concepts

The second approach uses statistical modelling as its central component while incorporating the linguistic concepts from the course.

Unit I – Morphology

Words may appear in different morphological forms.

For example:

Word	Stemmed Form
arrive	arriv
arrived	arriv
arriving	arriv
delivery	deliver
delivered	deliv
cancellation	cancel

Stemming reduces some surface-level variation before vectorization.

Regular expressions can additionally identify order numbers, dates and other structured information.

Unit II – N-Grams and POS

The classifier does not use only individual words.

For the sentence:

"I just want to know when it will arrive."

the vectorizer can generate:

Unigrams

want

know

arrive

Bigrams

want know

know when

will arrive

Trigrams

just want know

want know when

know when it

N-grams allow the model to learn combinations of words associated with specific intents.

For example:

cancel order

track order

where order

when arrive

payment failed

POS tags provide additional grammatical information that can be used during contextual correction.

Unit III – CFG and Negation

CFG parsing provides syntactic information about the customer message.

A simplified grammar can be:

S → NP VP

VP → V NP

VP → V AdvP

NP → Det N

Negation is particularly important.

For:

“I don't want to cancel it.”

the system identifies:

cancel → cancellation concept

don't want → negation

The statistical classifier may detect cancellation-related features, but the negation module prevents that evidence from being treated as a positive cancellation request.

Unit IV – Semantic Analysis

The system associates words and phrases with semantic concepts.

For example:

“hasn't arrived”

↓

Delivery status problem

↓

Order Tracking

A simplified FOPC representation can be:

Ordered(Customer, Laptop)

NotArrived(Order)

Wants(Customer, DeliveryInformation)

NotWants(Customer, Cancellation)

WSD is applied when words have multiple possible meanings.

Unit V – Discourse and Context

Discourse markers provide information that a bag-of-words representation cannot fully capture.

Marker	Interpretation
but	Contrast
still	Unresolved/continuing condition
just	Restriction or emphasis
it	Reference to earlier entity

The system therefore passes the classifier's prediction through a **contextual correction layer**.

5. Design / Proposed Solution / Methodology

The major difference in this version is that **statistical classification is responsible for the initial intent prediction**.

Customer Conversation

↓

Preprocessing

↓

Tokenization + Stemming

↓

Unigram / Bigram / Trigram Generation

↓

TF-IDF Vectorization

↓

Statistical Intent Classifier

↓

Initial Intent + Confidence

↓

Linguistic Analysis

↓

Negation Detection

↓

Entity + Reference Resolution

↓

Discourse Analysis

↓

Contextual Correction

↓

Final Intent

↓

Information Retrieval

↓

Response Generation

↓

Tamil Translation

The assignment's proposed NLP stages are therefore retained, but the **classification stage is implemented statistically rather than primarily through manually weighted rules**.

TF-IDF Representation

For a term $t$in document d:

$$TFIDF(t,d) = TF(t,d) \times IDF(t)$$

where:

$$IDF(t) = \log\left(\frac{N+1}{DF(t)+1}\right) + 1$$

The resulting vector represents how important terms and phrases are within the training corpus.

Classifier

A suitable classifier is **Logistic Regression**.

The classifier produces scores for all possible intents:

Order Tracking → 0.61

Delivery Information → 0.19

Order Cancellation → 0.13

Complaint → 0.04

Other → 0.03

The exact values above are illustrative; the actual implementation should report its own values.

The initial prediction is then checked against contextual information.

6. Algorithm / Pseudocode

```
BEGIN

Load labelled intent dataset

Preprocess training text
   tokenize
   normalize
   remove unnecessary symbols
   stem words

Create TF-IDF vectorizer
   ngram_range = (1,3)

Transform training sentences into TF-IDF vectors

Train statistical classifier

FOR each new conversation:

   preprocess conversation

   generate TF-IDF representation

   classifier_output ← predict probabilities

   initial_intent ← highest probability intent
```

```
perform POS tagging

perform CFG parsing

detect negation

extract entities

resolve references using previous turns

analyze discourse markers

IF initial_intent = Order Cancellation
  AND cancellation is negated:
     reduce cancellation confidence
     increase relevant alternative intent

IF delivery evidence exists:
     increase Order Tracking score

IF arrival/delivery request exists:
     increase delivery-related score

final_intent ← highest context-adjusted score

retrieve relevant information

generate response
```

translate response into Tamil

display all intermediate and final outputs

END

The pseudocode includes the functions, conditions, data processing and decision mechanism required by the assignment.

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7. Implementation / Source Code and Environment / Tools Used

Environment

Component	Tool
Language	Python
Development	Jupyter Notebook / VS Code
NLP	NLTK
Machine Learning	Scikit-learn
Vectorization	TfidfVectorizer
Classifier	Logistic Regression
Parser	NLTK CFG
Data Structures	Lists, dictionaries, arrays
Translation	Suitable translation library/API

Main Implementation Modules

load_dataset()

preprocess_text()

stem_text()

build_tfidf()

train_classifier()

predict_intent()

perform_pos_analysis()

detect_negation()

extract_entities()

resolve_reference()

analyze_discourse()

adjust_prediction()

retrieve_information()

generate_response()

translate_response()

Example TF-IDF Configuration

```
TfidfVectorizer(
    ngram_range=(1,3),
    min_df=1,
    sublinear_tf=True
)
```

The important part is the use of **1–3 grams**, since the assignment explicitly requires unigram, bigram and trigram representations.

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8. Test Cases and Expected / Actual Results

Test Case	Customer Input	Expected Intent
TC1	“Where can I track my order?”	Order Tracking
TC2	“Please cancel my order.”	Order Cancellation
TC3	“My package has not arrived.”	Order Tracking
TC4	“I need to return this item.”	Return Request
TC5	“The payment was unsuccessful.”	Payment Problem
TC6	“I don't want to cancel it, I want to know when it will arrive.”	Order Tracking

Test Case	Customer Input	Expected Intent
TC7	“What features does this laptop have?”	Product Information

Main Conversation Test

Input:

“I ordered a laptop three days ago, but it still hasn't arrived. Can you check where it is?”

Followed by:

“I don't want to cancel it. I just want to know when it will arrive.”

Expected:

Initial classifier prediction:

Delivery-related intent

Contextual analysis:

Cancellation → negated

Final intent:

Order Tracking / Delivery Information

The actual probability values should be obtained from the trained classifier.

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9. Execution Screenshots / Outputs

The following screenshots should be captured from the actual implementation.

Screenshot 1 – Preprocessing/tokenization output.

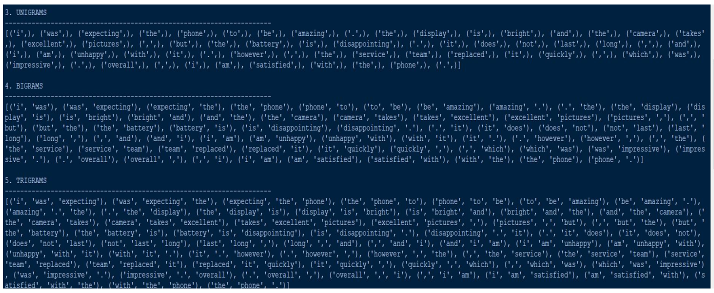
```
1. PREPROCESSING / TOKENIZATION
----------------------------------------------------------------------
['i', 'was', 'expecting', 'the', 'phone', 'to', 'be', 'amazing', '.', 'the', 'display', 'is', 'bright', 'and', 'the', 'camera', 'takes', 'excellent', 'pictures', ',', 'but', 'the', 'ba
ttery', 'is', 'disappointing', '.', 'it', 'does', 'not', 'last', 'long', ',', 'and', 'i', 'am', 'unhappy', 'with', 'it', '.', 'however', ',', 'the', 'service', 'team', 'replaced', 'it'
, 'quickly', ',', 'which', 'was', 'impressive', '.', 'overall', ',', 'i', 'am', 'satisfied', 'with', 'the', 'phone', '.']
```

User:

I ordered a laptop three days ago,

but it still hasn't arrived.

Screenshot 2 – N-gram and POS-tagging output screenshot.

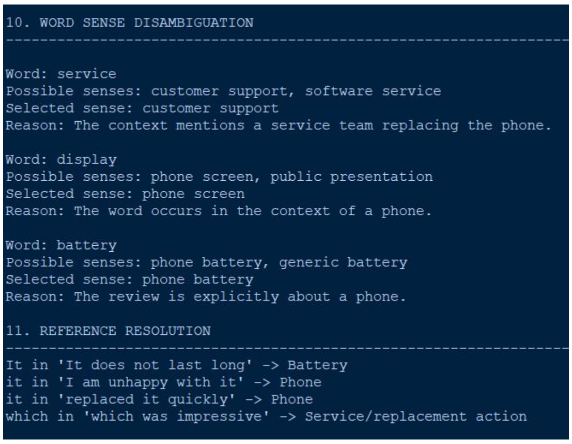


```
3. UNIGRAMS
--------------------------------------------------------------------------
[('i',), ('was',), ('expecting',), ('the',), ('phone',), ('to',), ('be',), ('amazing',), ('.',), ('the',), ('display',), ('is',), ('bright',), ('and',), ('the',), ('camera',), ('takes'
,), ('excellent',), ('pictures',), (',',), ('but',), ('the',), ('battery',), ('is',), ('disappointing',), ('.',), ('it',), ('does',), ('not',), ('last',), ('long',), (',',), ('and',),
('i',), ('am',), ('unhappy',), ('with',), ('it',), ('.',), ('however',), (',',), ('the',), ('service',), ('team',), ('replaced',), ('it',), ('quickly',), (',',), ('which',), ('was',),
('impressive',), ('.',), ('overall',), (',',), ('i',), ('am',), ('satisfied',), ('with',), ('the',), ('phone',), ('.',)]

4. BIGRAMS
--------------------------------------------------------------------------
[('i', 'was'), ('was', 'expecting'), ('expecting', 'the'), ('the', 'phone'), ('phone', 'to'), ('to', 'be'), ('be', 'amazing'), ('amazing', '.'), ('.', 'the'), ('the', 'display'), ('dis
play', 'is'), ('is', 'bright'), ('bright', 'and'), ('and', 'the'), ('the', 'camera'), ('camera', 'takes'), ('takes', 'excellent'), ('excellent', 'pictures'), ('pictures', ','), (',', '
but'), ('but', 'the'), ('the', 'battery'), ('battery', 'is'), ('is', 'disappointing'), ('disappointing', '.'), ('.', 'it'), ('it', 'does'), ('does', 'not'), ('not', 'last'), ('last', '
long'), ('long', ','), (',', 'and'), ('and', 'i'), ('i', 'am'), ('am', 'unhappy'), ('unhappy', 'with'), ('with', 'it'), ('it', '.'), ('.', 'however'), ('however', ','), (',', 'the'), (
'the', 'service'), ('service', 'team'), ('team', 'replaced'), ('replaced', 'it'), ('it', 'quickly'), ('quickly', ','), (',', 'which'), ('which', 'was'), ('was', 'impressive'), ('impres
sive', '.'), ('.', 'overall'), ('overall', ','), (',', 'i'), ('i', 'am'), ('am', 'satisfied'), ('satisfied', 'with'), ('with', 'the'), ('the', 'phone'), ('phone', '.')]

5. TRIGRAMS
--------------------------------------------------------------------------
[('i', 'was', 'expecting'), ('was', 'expecting', 'the'), ('expecting', 'the', 'phone'), ('the', 'phone', 'to'), ('phone', 'to', 'be'), ('to', 'be', 'amazing'), ('be', 'amazing', '.'),
('amazing', '.', 'the'), ('.', 'the', 'display'), ('the', 'display', 'is'), ('display', 'is', 'bright'), ('is', 'bright', 'and'), ('bright', 'and', 'the'), ('and', 'the', 'camera'), ('
the', 'camera', 'takes'), ('camera', 'takes', 'excellent'), ('takes', 'excellent', 'pictures'), ('excellent', 'pictures', ','), ('pictures', ',', 'but'), (',', 'but', 'the'), ('but', '
the', 'battery'), ('the', 'battery', 'is'), ('battery', 'is', 'disappointing'), ('is', 'disappointing', '.'), ('disappointing', '.', 'it'), ('.', 'it', 'does'), ('it', 'does', 'not'),
('does', 'not', 'last'), ('not', 'last', 'long'), ('last', 'long', ','), ('long', ',', 'and'), (',', 'and', 'i'), ('and', 'i', 'am'), ('i', 'am', 'unhappy'), ('am', 'unhappy', 'with'),
('unhappy', 'with', 'it'), ('with', 'it', '.'), ('it', '.', 'however'), ('.', 'however', ','), ('however', ',', 'the'), (',', 'the', 'service'), ('the', 'service', 'team'), ('service',
'team', 'replaced'), ('team', 'replaced', 'it'), ('replaced', 'it', 'quickly'), ('it', 'quickly', ','), ('quickly', ',', 'which'), (',', 'which', 'was'), ('which', 'was', 'impressive')
, ('was', 'impressive', '.'), ('impressive', '.', 'overall'), ('.', 'overall', ','), ('overall', ',', 'i'), (',', 'i', 'am'), ('i', 'am', 'satisfied'), ('am', 'satisfied', 'with'), ('s
atisfied', 'with', 'the'), ('with', 'the', 'phone'), ('the', 'phone', '.')]
```

Screenshot 3 – Reference Resolution



```
10. WORD SENSE DISAMBIGUATION
------------------------------------------------------------------

Word: service
Possible senses: customer support, software service
Selected sense: customer support
Reason: The context mentions a service team replacing the phone.

Word: display
Possible senses: phone screen, public presentation
Selected sense: phone screen
Reason: The word occurs in the context of a phone.

Word: battery
Possible senses: phone battery, generic battery
Selected sense: phone battery
Reason: The review is explicitly about a phone.

11. REFERENCE RESOLUTION
------------------------------------------------------------------
It in 'It does not last long' -> Battery
it in 'I am unhappy with it' -> Phone
it in 'replaced it quickly' -> Phone
which in 'which was impressive' -> Service/replacement action
```

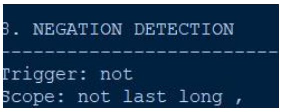
Reference:

"it"

Resolved To:

Existing Order

Screenshot 4 – Negation Analysis



```
8. NEGATION DETECTION
-------------------------
Trigger: not
Scope: not last long ,
```

Detected Action:

Cancellation

Negation:

TRUE

Interpretation:

Customer rejects cancellation

Screenshot 5 – Final State

Conversation State

Product : Laptop

Order Status : Existing

Delivery Status : Not Received

Cancellation : Rejected

Current Reference : Order

Requested Information: Arrival Time

Current Intent : Order Tracking

Screenshot 6 – Response

Final Intent:

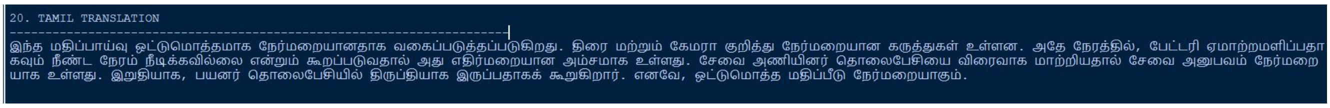
Order Tracking

Response:

Please provide your order number so I can check

the current delivery status of your laptop.

Tamil Response:



The assignment requires evidence of intermediate NLP representations, final classification and generated chatbot output.

Response:

Please provide your order number so I can check

the current delivery status.

Tamil Response:

[translated response]

The assignment requires evidence including program outputs, screenshots, intermediate representations, final classification and generated responses.

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10. Results and Validation

Morphological Analysis

Word	Stem
ordered	order
arrived	arriv
arriving	arriv
delivery	deliver
delivered	deliv
cancelling	cancel

The statistical representation benefits from normalization because multiple surface forms can contribute to related vocabulary patterns.

Intent and Entity Analysis

Component	Result	Evidence
Product	Laptop	“ordered a laptop”
Order	Existing order	“I ordered...”
Time	Three days	“three days ago”
Delivery	Delayed	“hasn't arrived”
Cancellation	Negated	“don't want to cancel”
Reference	“it” → order	Previous message
Final Intent	Order Tracking	Arrival information requested

Classifier Validation

The actual implementation should calculate metrics such as:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion matrix

For example:

Metric	Actual Result
Accuracy	[actual value]
Precision	[actual value]
Recall	[actual value]
F1-score	[actual value]

These values should **not be fabricated** and must be filled using the student's actual test execution.

The assignment expects the system to validate different expressions, morphological variations, n-grams, negation, ambiguity, references, conversational context and intent classification.

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11. Analysis, Comparison, Trade-offs and Justification

The major strength of the statistical approach is its ability to learn patterns from examples.

Consider these expressions:

Where is my order?

When will my package arrive?

My parcel hasn't arrived.

Can you tell me the delivery status?

When will my item be delivered?

A keyword-only system may treat these as different because the vocabulary changes.

A TF-IDF n-gram classifier can learn that terms and phrases occurring across labelled delivery-related examples are associated with the same intent.

Morphological Variation

Words such as:

arrive

arrived

arriving

arrival

delivery

delivered

represent related concepts but are not identical strings.

Preprocessing and statistical representation help the classifier generalize across such variations.

Why N-Grams Matter

The word **cancel** alone could indicate cancellation.

However:

want cancel

don't want cancel

order cancellation

cancel my order

have different meanings.

Therefore, surrounding terms provide useful classification information.

Why the Classifier Alone Is Not Enough

The statistical model may still incorrectly respond to the strong cancellation signal in:

“I don't want to cancel it.”

This is why the proposed system includes a linguistic correction layer.

Comparison

Method	Strength	Weakness
Keyword matching	Simple and fast	Poor contextual understanding
TF-IDF + classifier	Learns statistical patterns	Can struggle with negation
Rule-based system	Explainable	Requires extensive manual rules
Proposed hybrid statistical approach	Learns patterns + handles linguistic exceptions	More complex to implement

The final approach is justified because it combines **data-driven classification with explicit linguistic correction**.

12. Broader Considerations / SDG Relevance

The assignment maps the application to **SDG 9 – Industry, Innovation and Infrastructure**, particularly because automated customer-service systems represent the use of intelligent digital technologies in industry.

Privacy

Customer conversations can contain sensitive information such as order numbers, addresses and payment details. Such information should not be unnecessarily stored or exposed.

Bias

A classifier trained on limited examples may perform poorly on uncommon sentence structures, spelling variations or expressions not represented in the training data.

Incorrect Classification

A wrong prediction could lead to an inappropriate customer-service response. This is especially important for actions such as cancellation or refund processing.

Multilingual Accessibility

Tamil translation can make the chatbot more accessible to customers who prefer Tamil.

Responsible AI

The chatbot should clearly distinguish between retrieving information, recommending an action and actually performing that action.

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13. Conclusion, Limitations and Possible Improvements

Conclusion

A statistical NLP approach was designed for e-commerce chatbot intent classification. The system transforms customer conversations into TF-IDF representations containing unigram, bigram and trigram features and uses a statistical classifier to generate an initial intent prediction.

However, classification is not allowed to depend entirely on statistical word patterns. Therefore, the predicted intent is passed through additional linguistic processing involving negation, entities, references and discourse.

For the given conversation, the final prediction is **Order Tracking / Delivery Information**. Although the classifier may detect cancellation-related vocabulary, the contextual correction layer recognizes that cancellation has been explicitly rejected.

This demonstrates why combining statistical NLP with contextual linguistic analysis provides a more reliable solution than simple keyword matching.

Limitations

- Classification performance depends on training-data quality.
- Rare intents may have insufficient examples.
- TF-IDF does not fully represent deep semantic relationships.
- Negation may still be difficult in complex sentences.
- Reference resolution becomes harder with long conversations.
- Translation quality depends on the external translation system.

Possible Improvements

Future work could include:

- Word embeddings instead of only TF-IDF.
- Transformer-based intent classification.
- Larger and more diverse conversational datasets.
- Advanced coreference resolution.
- Multilingual training data.
- Real-time integration with e-commerce databases.
- Confidence thresholds for transferring difficult cases to human agents.

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14. Individual Contribution of Group Members

The following should be replaced with the **actual work performed by the group members**.

Member	Contribution
Harissh M	Dataset preparation and preprocessing
Pradeep Kumar S	TF-IDF and n-gram implementation
Lokanath reddy	Classifier development and evaluation
Somashekhar naidu	Negation, reference and contextual correction
G. Jaswanth	Response generation, translation and documentation
Yuvaraj R	Testing & Validation

The contribution table should reflect actual work rather than assigning tasks retrospectively.

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15. References

The report should cite the actual resources used during development. The assignment specifically requires references to relevant textbooks, research papers, NLP libraries, datasets, software/tools, websites and AI-assisted tools where applicable.

Possible reference categories include:

1. NLP textbook used for the course.
2. NLTK documentation.
3. Scikit-learn documentation.
4. Python documentation.
5. Research papers related to intent classification.
6. Dataset documentation.
7. Translation library/API documentation.
8. Any AI-assisted development tools actually used.

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16. One-Page Individual Reflection

Individual Reflection

Module 5: Response Generation, Translation and Documentation

This assignment gave me an opportunity to understand how an NLP system can convert its final intent classification into a meaningful response for the customer. Before working on this project, I mainly focused on the prediction stage and considered classification as the final output. During implementation, I learned that an NLP chatbot must also retrieve relevant information, generate an appropriate response and present it in a form that is understandable to the user.

My main contribution to the project was the **response generation, translation and documentation module**. I worked on designing the response generation process based on the final intent and the information identified from the conversation. The system uses the final contextual intent, such as Order Tracking or Delivery Information, to select an appropriate response. This helped me understand how the output of NLP analysis can be connected to a practical conversational application.

I also worked on the **Tamil translation component** of the system. The generated English response is passed to a translation stage so that the chatbot can provide the response in Tamil. Through this implementation, I learned that multilingual NLP is not simply a matter of replacing individual words. The meaning and structure of the complete response need to be preserved during translation. I also understood that the quality of the final output depends on the translation model or API being used.

One of the challenging aspects of this module was ensuring that the generated response was consistent with the final intent and the conversation context. For example, when the customer says, *“I don't want to cancel it. I just want to know when it will arrive,”* the system should not generate a cancellation response merely because the word “cancel” appears. The response must reflect the contextual correction and provide delivery-related information instead.

I was also involved in documenting the overall system workflow, intermediate outputs and implementation details. This helped me understand the importance of technical documentation in an NLP project because the final system involves multiple stages such as preprocessing, TF-IDF representation, statistical classification, linguistic analysis, contextual correction, information retrieval, response generation and translation.

Overall, this assignment improved my understanding of how an NLP classification system can be transformed into a complete conversational application. I learned that accurate intent prediction is only one part of a chatbot, while response generation, multilingual accessibility and clear documentation are equally important for making the system useful to real users.